Chapter 2: Vector Models and Text Preprocessing

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Outline

1. Basic Definitions in NLP
2. Vector Representations
3. Bag of Words Representation
4. Count Vectorizer

Basic Definitions in NLP

Tokens and Tokenization

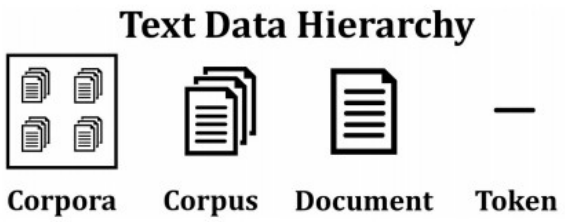
- Tokens are basic units (words, punctuation, etc.).
- Tokenization splits text into these units.

Vocabulary in NLP

- The vocabulary is the set of words used.
- Often a subset is chosen for efficiency.
- Can range from thousands to millions of words.

What is a Corpus?

- A corpus is a large collection of texts.
- Examples: articles, tweets, books.
- A well-curated corpus boosts model performance.



N-Grams in NLP

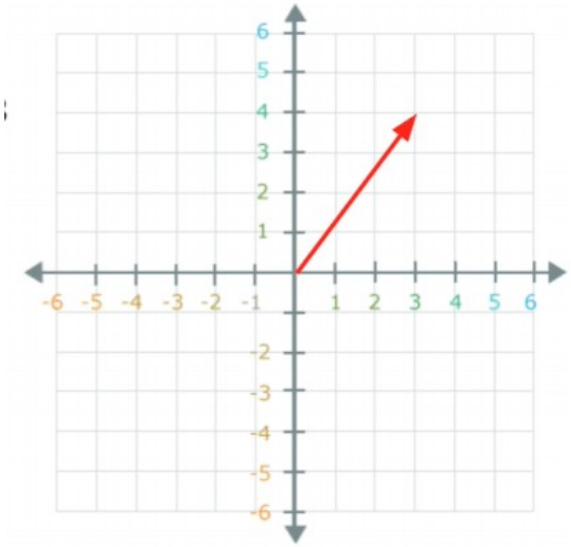
- Sequences of N consecutive words.
- Unigram: 1 word, Bigram: 2 words, Trigram: 3 words.
- Useful in language models.

Text	N-gram
Data	1-gram
Great information	2-gram
I am fine	3-gram
Nice to meet you	4-gram

Vector Representations

What is a Vector?

- A numerical representation of text.
- The basis of many ML algorithms.

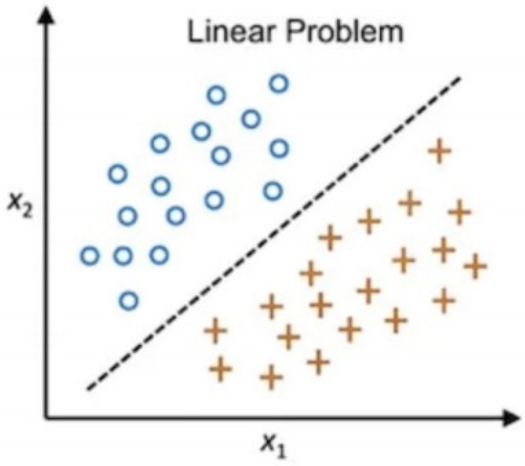


Why are Vectors Useful?

- They allow nuanced, context-based comparisons.
- Increase the robustness of ML models.
- Essential for sentiment analysis, translation, etc.

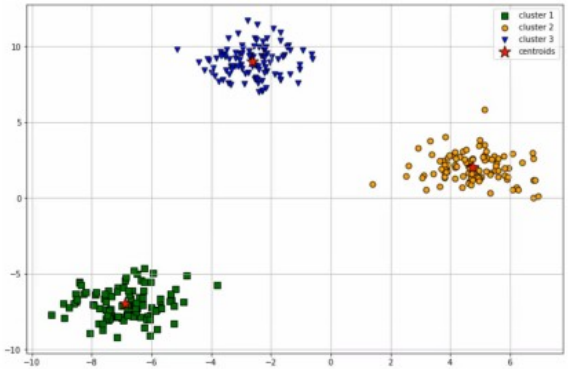
Spam Detection with Vectors

- Emails are converted into vectors.
- Spam and non-spam form distinct clusters.
- A new email's position determines its class.



Organizing Documents Using Vectors

- Enables clustering of similar documents.
- Useful for search engines and organization.



Challenges in Vector Representations

- Good vectors simplify ML tasks.
- Poor vectors cause overlapping clusters.
- Effective vectorization techniques are essential.

Bag of Words Representation

Bag of Words

- Ignores word order.
- Only word frequency is counted.
- A simple and effective method.

Where is Bag of Words Used?

- Common in classical ML (classification, sentiment analysis).
- Less so in deep learning or sequential tasks.
- Sometimes used in hybrid models.

Count Vectorizer

Count Vectorizer

- Converts text into a count vector.
- Uses the BoW approach.
- Simple to implement and interpret.

Documents

- A document can be an article, chapter, or a sentence.
- Each document is one data sample.

Data Format

- Data is stored in CSV, Excel, or DataFrames.
- Each row contains the text and its label.

Determining Vocabulary Size

- V is the number of unique words.
- Each document is converted into a vector of dimension V.

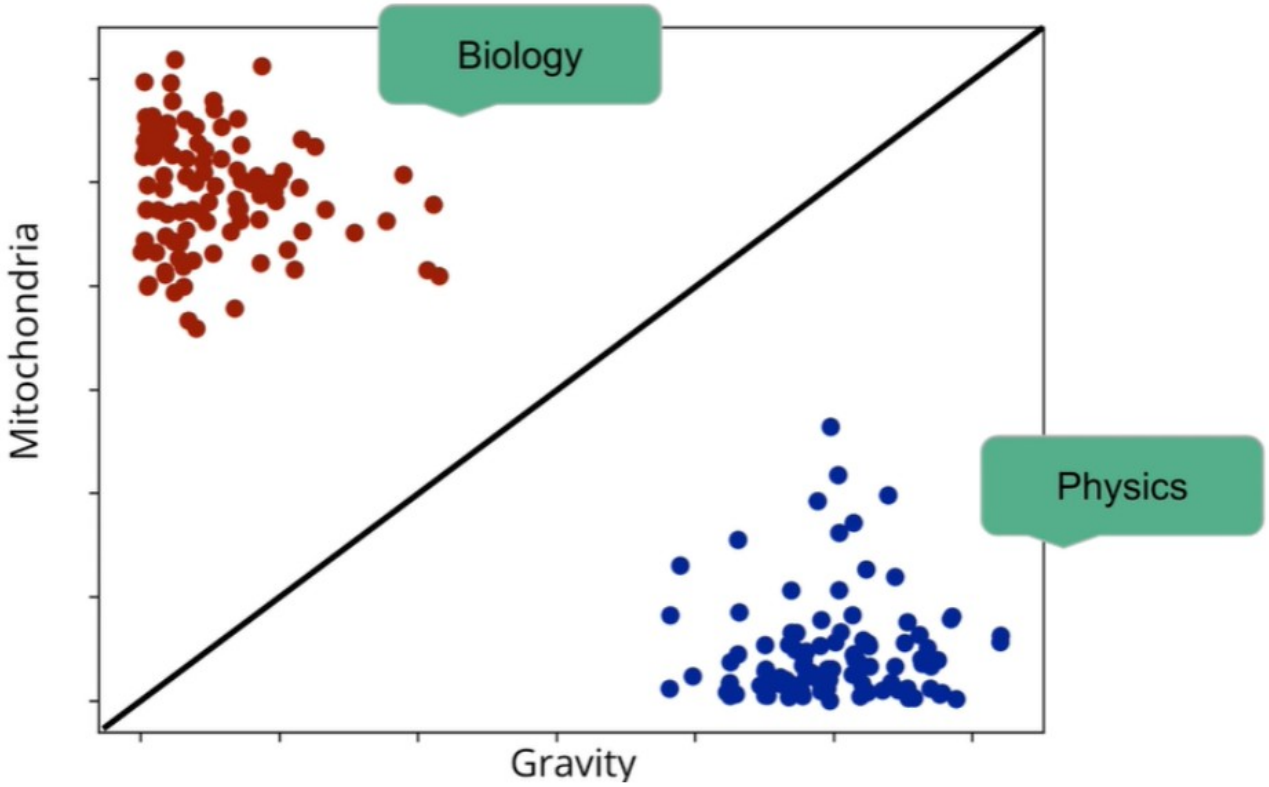
This means "and" appears in the document 25 times

An example vector

0	0	25	...	0	3
"a"	"aa"	"and"	...	"zoo"	"zygote"

How Does Counting Help?

- Frequencies help differentiate topics.
- E.g., "cell" in biology vs. "gravity" in physics.



Count Vectorizer in Code (Scikit-Learn)

- Three simple steps:
  1. Initialize `CountVectorizer`.
  2. `fit` on the training corpus.
  3. `transform` the documents.

```
vectorizer = CountVectorizer()
Xtrain = vectorizer.fit_transform(list_of_documents_train)
Xtest = vectorizer.transform(list_of_documents_test)
```

Normalization in Text Vectorization

- **L2 Normalization:**

$$\hat{x} = \frac{x}{\|x\|_2} \quad (\|x\|_2 = \sqrt{\sum x_i^2})$$

- Scales the vector so its norm is 1.

Normalization in Text Vectorization

- **L1 Normalization:**

$$\hat{x} = \frac{x}{\|x\|_1} \quad (\|x\|_1 = \sum |x_i|)$$

- The sum of components equals 1.
- `CountVectorizer` does not normalize by default.